SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

60.002 A.I. Applications and Design

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1 Intro

1.1 Project Context

This project is conducted under the mandate of a consulting firm tasked with identifying substantive design improvement opportunities through automated, data-driven text analytics. In an era where consumer feedback is decentralized across various digital retail platforms, traditional manual market research is often slow and prone to human bias. This initiative utilizes an end-to-end AI pipeline to process unstructured digital sentiment, enabling a more agile and objective approach to identifying market gaps and user pain points.

1.2 Product Selection & Importance

For this study, our team has selected the toilet bowl as the primary subject of analysis. While often viewed as a static utility, the toilet bowl is a critical fixture in global sanitation and public health. We selected this product for several strategic reasons:

- Despite its fundamental necessity, users consistently report issues with flush reliability, cleaning difficulty, and ergonomic comfort, indicating persistent design gaps.
- Optimizing toilet design is directly linked to Sustainable Development Goal 6 (Clean Water and Sanitation). In the context of Singapore's Smart Nation goals, even marginal improvements in water efficiency and durability can have significant cumulative impacts.
- The high volume of technical and lifestyle-oriented reviews across diverse retailers (e.g., Home Depot, Amazon) makes the toilet bowl an ideal candidate for benchmarking current hardware against user expectations.

1.3 Objective

The primary objective of this project is to utilize a systematic AI-driven workflow to identify design opportunities and formalize measurable product requirements. Crucially, the focus remains on the identification of what needs to be improved based on real-world evidence, rather than the premature generation of engineering solutions.

Beyond the toilet bowl, a key goal of this project is to demonstrate the versatility of the developed framework. The Python-based pipeline is designed to be product-agnostic; it is a scalable consulting tool that can be redeployed for any product category, from consumer electronics to industrial tools, simply by initializing the system with a new product query. This ensures that the methodology remains a high-value, repeatable asset for the firm.

2 Methodology

2.1 System Architecture

Our consulting framework utilizes an automated 6-stage pipeline to transform unstructured consumer sentiment into prioritized engineering specifications. This workflow is designed to identify design opportunities with high granularity while eliminating manual human bias.

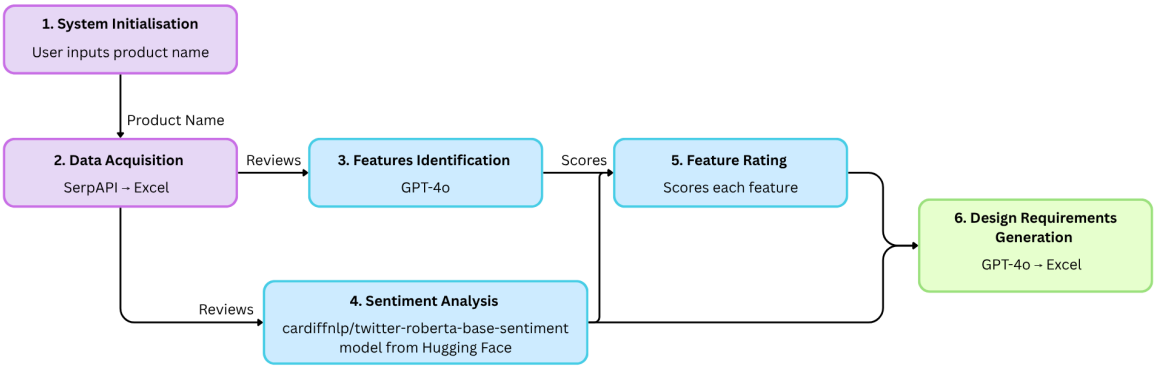

Figure 1: Design Requirements Generation Pipeline

Stage 1: System Initialisation

- The pipeline initializes by capturing the target product name (e.g. "toilet bowl") to focus the data acquisition.
- A dedicated directory is automatically established in Google Drive for version control and full auditability of all raw and processed outputs.

Stage 2: Data Acquisition

- The pipeline then utilises SerpAPI to query Google Shopping and aggregate up to 50 real-world user reviews from diverse retail sources.
- Captured data is then exported to an Excel format (01_raw_reviews.xlsx) to establish a verifiable baseline.

Stage 3: Features Identification

- GPT-4o is leveraged to generate an initial list of 8–10 key product features that users typically care about when reviewing toilet bowls, such as flush efficiency, comfort height, and ease of cleaning. These features are then validated through keyword matching and mention counts in the collected reviews.

Stage 4: Sentiment Analysis

- The system employs the *cardiffnlp/twitter-roberta-base-sentiment* model from HuggingFace to perform a sentence-level analysis.

Stage 5: Feature Rating

- The pipeline matches isolated sentences to identified features and averages the qualitative sentiment scores.
- These scores are mathematically mapped to a 1–5 star rating system, providing an immediate visual benchmark of design strengths and weaknesses.

Stage 6: Design Requirements Generation

- The final stage feeds the aggregated sentiment summary and sample reviews into GPT-4o to synthesize measurable product requirements.
- The AI is prompted to act as a product designer, translating vague user complaints into structured, measurable requirements.

- All analytical sheets are consolidated into a master Excel file (product_analysis.xlsx), providing a centralized 'source of truth' for design decision-making.

2.2 Justification

2.2.1 API-Driven Acquisition vs Manual Scraping

We utilised SerpAPI for data acquisition to ensure a legally compliant and technically stable bridge to retail data. Unlike traditional web scraping, which is prone to failure due to UI updates and bot-blocking measures, an API-first approach ensures the integrity, repeatability and legal safety of the consulting process.

2.2.2 Data-Driven Feature Extraction

By utilizing GPT-4o in Stage 3, we reduced consultant bias in the early feature selection process. Instead of manually deciding the features from the outset, GPT-4o generated an initial feature list based on common user concerns for toilet bowls. The relevance of these features was then assessed using review-level mention counts during sentiment analysis.

2.2.3 Granular Sentence-Level Processing

A critical pillar of our methodology is the shift from review-level to sentence-level analysis. The model can then capture nuanced feedback, such as a user praising the aesthetic design while simultaneously criticizing the flush power. By isolating feature-specific sentences, we capture these design opportunities that generalized star ratings often miss.

2.2.4 Bridging the Data-to-Design Gap

The final transition in Stage 6 converts raw data into measurable engineering specifications. This bridges the gap between how users feel and what designers must build, providing the firm with actionable, data-driven recommendations rather than vague insights.

3 Data Collection

3.1 Data Acquisition Strategy

The primary data source for this project is Google Shopping, accessed programmatically via SerpAPI. This platform was selected because it serves as a high-authority aggregator, surfacing product reviews from a wide array of major retailers including Amazon, Home Depot, Lowe's, and specialty plumbing outlets.

A critical strategic decision was made to avoid direct web scraping of Amazon for the following reasons:

- **Legal & Ethical Compliance**
  - Direct scraping often violates the Terms of Service (ToS) of major retailers.
  - SerpAPI provides a legally compliant middleware that respects robots.txt and site-specific crawling policies.
- **Technical Stability**
  - Amazon employs sophisticated anti-bot measures (CAPTCHAs, IP blocking) and frequent UI changes that make custom scrapers brittle.
  - SerpAPI provides a structured, JSON-based interface that ensures data integrity and pipeline reliability.
- **Data Aggregation**

    - While Amazon is a massive retailer, Google Shopping provides a more holistic market view by aggregating reviews from multiple competitors, which is essential for a comprehensive consulting benchmark.

3.2 Dataset Composition

The pipeline successfully captured a target sample of 50 reviews for the toilet bowl category. For each entry, we extracted the following standardized data points to facilitate downstream analysis:

- **Review Text Content:** The primary qualitative data used for sentiment extraction.
- **Numerical Ratings:** 1–5 star ratings used to correlate sentiment with user satisfaction.
- **Source Attribution:** Identifying the original retailer (e.g., "Verified purchase from Home Depot") to assess source authority.
- **Review Date:** To ensure the analysis focuses on current product iterations rather than discontinued models.

3.3 Data Diversity

To ensure a robust breadth of analysis, we evaluated the composition of our dataset:

- **Source Diversity**
    - The dataset represents a healthy mix of Big-box retailers (Home Depot, Lowe's) and e-commerce giants (Amazon).
    - This diversity is vital because user expectations can vary by platform (e.g., professional contractors on Home Depot vs. DIY homeowners on Amazon).
- **Information Density**
    - We observed that professional plumbing site reviews tend to be more technical, focusing on rough-in dimensions), while Amazon reviews are more lifestyle-oriented (focusing on comfort and aesthetics). Our pipeline effectively captured both perspectives.

4 Feature Definition

4.1 How Features Were Selected

The features analysed based on the ‘Toilet Bowl’ product category includes 8-10 key aspects that were extracted by the GPT-4o model itself based on our prompt:

```
USE_AUTO_FEATURES = True

if USE_AUTO_FEATURES:
        from openai import OpenAI
        client = OpenAI(api_key=openai_key)
        prompt = f"""List 8-10 key features or aspects that users typically care about when reviewing a '{product_name}'. Return ONLY a Python list of short lowercase strings, like: [\"battery life\", \"sound quality\", ...]"""

        response = client.chat.completions.create(
        model='gpt-4o',
        messages=[{'role': 'user', 'content': prompt}]
        )
```

Figure 2: GPT-4o feature extraction prompt and API call

This automation helps to ensure that the selected features are grounded in actual user concerns and not based on what we ourselves define as key aspects in the product, preventing our use of predefined assumptions and reducing subjective bias in the early-stage of the design analysis.

The phrase 'users typically care about when reviewing a '{product_name}' grounds GPT-4o model into the perspective of real consumers rather than engineering specifications, avoiding generations of technical parameters that users would never discuss about the product.

We restricted the features to 8-10 in order to ensure a manageable number of features to analyse as too few would lead to a surface-level analysis whereas too many would lead to dilution of sentiment data across too many categories which decreases the production of statistically meaningful data/scores.

The instruction of returning only a Python list of lowercase strings ensures keyword matching later on in the further code blocks where reviews will be converted into lowercase before comparison. This helps to ensure that these operate consistently while avoiding any mismatch due to capitalisation differences.

Although our code system does support manual feature definition inputs by us, we decided to automate it instead to maintain objectivity and consistency across the product's categories. Manual input of features can still be done by directly modifying the features array (features = []) and setting USE_AUTO_FEATURES = False.

4.2 Features Analysed

As of writing this report, the most recent and relevant key aspects to be analysed of the product are as mentioned (with no order of priority yet):
1. Flush Efficiency
2. Water Usage
3. Comfort Height
4. Bowl Shape
5. Ease of Cleaning
6. Installation Process
7. Noise Level
8. Durability
9. Design Aesthetics
10. Price

These features are based on GPT-4o's general understanding of common user concerns for toilet bowls. Their relevance to our specific dataset was then assessed through keyword matching, feature mention counts, and sentiment scores.

5 Sentiment Analysis

5.1 Model Overview

The pipeline utilises the HuggingFace transformer model: cardiffnlp/twitter-roberta-base-sentiment, a RoBERTa (Robustly Optimised BERT Pretraining Approach) deep-learning model used for our sentiment classification. Its purpose would be to programmatically evaluate the emotional valence of positive, neutral or negative through consumer-written text at the sentence level and attribute it to the respective product features identified in Stage 3 (feature analysis). The model has been pre-trained on

58 million tweets, fine-tuned specifically on a massive scale of social media texts, closely mirroring the informal, short, informal and emotionally expressive writing style found in most consumer product reviews where it produces confidence-weighted scores based on the label mapping of:

```
LABEL_MAP = {'LABEL_0': -1, 'LABEL_1': 0, 'LABEL_2': 1}
```

5.1.1 Sentence-Level Analysis - Sentence Segmentation

Unlike other natural language processing (NLP) transformers, RoBERTa applies sentence segmentation logic and feature-keyword matching into a single unified analytic function. Rather than analysing entire reviews as a single document reading text left-to-right, the pipeline splits each review into individual sentences and processes the entire sentence simultaneously in both directions, enabling it to understand how words modify each other's meaning across the full attention span. This helps to capture mixed feedback where a user might praise flush efficiency in one sentence and criticise noise level in another. Transformer-based models can capture contextual negation and modifier effects that other simpler approaches may systematically classify.

```
def split_sentences(text):
    return [s.strip() for s in re.split(r'[.!?]', str(text)) if len(s.strip()) > 10]
```

Figure 3: Sentence Splitting function

The function in Figure 3 breaks sentence-terminating punctuation (., !, ?) and filters out fragments shorter than 10 characters to remove any noise.

5.1.2 Feature Keyword Matching

For each sentence, the pipeline checks whether it contains any keywords from the feature list in 4.2 above. If a match is found, it passes the sentence to the sentiment model.

```
    sentence_lower = sentence.lower()
    for feature in features:
        feature_keywords = feature.split()
        if any(kw in sentence_lower for kw in feature_keywords):
            score = score_sentiment(sentence)
            feature_scores[feature].append(score)
```

Figure 4: Feature Matching logic

This logic is under the analyse_features(reviews_df, features) function where it splits the feature strings into individual keywords (e.g., "ease of cleaning" into ["ease", "of", "cleaning"]) and a sentence is considered a match if any keyword appears in the lowercased sentence text.

5.1.3 Sentiment Scoring and Storage

For each feature, all matched sentence scores are averaged to produce a mean sentiment score s (range: -1 to +1) which is then converted to a 1-5 star rating using the formula (further analysis can be found in section 6: Feature Ratings): Star Rating = ((s + 1) / 2) × 4 + 1

This linearly maps -1 to 1 star and +1 to 5 stars.

```
if len(feature_mentions[feature]) < 5:
        feature_mentions[feature].append(sentence.strip())
```

Figure 5: Storing Example Sentences (<= 5)

This logic is under the analyse_features(reviews_df, features). The returned confidence-weighted score is then appended to the feature's score list as seen in Figure 5. If <5 example sentences have been stored for that feature, the sentence is added to the feature_mentions list as a qualitative reference for us (human) to review.

5.2 Sentiment Analysis Results

Feature	Mentions	Star Rating	% Positive	% Negative
Comfort Height	7	4.2	85.7	0.0
Flush Efficiency	18	4.1	72.2	0.0
Design Aesthetics	5	4.1	60.0	0.0
Price	2	3.9	50.0	0.0
Ease of Cleaning	39	3.8	53.8	5.1
Bowl Shape	26	3.8	61.5	3.8
Water Usage	9	3.7	44.4	0.0
Installation Process	2	3.0	0.0	0.0
Noise Level	4	3.0	0.0	0.0
Durability	0	3.0	0.0	0.0

Figure 6: Sentiment Analysis Output

The mention count data in Figure 6 shows the relationship between feature coverage and analytical confidence. Features with high mention counts produce statistically robust average sentiment scores as they are computed from a larger sample pool whereas features with low mention counts produce scores based on fewer observations where a single sentence can dramatically skew the results.

The zero mentions for 'durability' across 47 reviews shows that users rarely comment on long-term reliability in short-term purchase reviews. For reviews with low mentions such as installation process, noise level and durability, their sentiment scores are statistically unreliable. However, we will still take these into design considerations for future design recommendations but with caution and note their low statistical confidence as a fundamental limitation of our analysis.

6 Feature Ratings

6.1 How Sentiment is Converted to a 1–5 Score

The sentiment analysis model (cardiffnlp/twitter-roberta-base-sentiment) outputs a label and a confidence score for each sentence. The label is mapped to a numeric value: Negative (LABEL_0) → −1, Neutral (LABEL_1) → 0, and Positive (LABEL_2) → +1. This raw score is then weighted by the model's confidence, producing a sentiment score between −1 and +1 for each sentence.

Step 1: Confidence-weighted sentence score:

$$Sentence\ Score = label\ value \times confidence$$

where label value $\in \{-1,0,+1\}$ and confidence $\in [0, 1]$

To make these scores more intuitive, they are converted into a 1–5 star rating using the formula:
Step 2 — Star rating conversion:

$$Star\ Rating\ =\ (\frac{\bar{s}+1}{2})\ \times\ 4\ +\ 1$$

$$where\ \bar{s} = \frac{1}{n}\sum_{i=1}^{n} Sentence\ Score_i$$

Where:
$\bar{s}\ =\ mean\ sentiment\ score\ across\ all\ matched\ sentences\ for\ a\ feature$
$n\ =\ total\ number\ of\ matched\ sentences\ for\ that\ feature$
$The\ output\ maps\ \ -\ 1{\rightarrow}1\bigstar,\ 0{\rightarrow}3\bigstar\ and\ \ +\ 1{\rightarrow}5\bigstar$

For each feature, all relevant sentence scores are averaged to produce a single representative star rating. The summary also tracks the percentage of positive and negative mentions, giving a fuller picture beyond just the average.

6.2 Averaging Method

The pipeline computes a per-feature average across all matched sentences from all 50 reviews. Rather than treating each review as one data point, it treats each relevant sentence as a data point — this means features that are discussed frequently are represented more robustly, while rarely-mentioned features carry more uncertainty.

6.3 Analysis: Which Features Scored High/Low

Since the product analysed was a toilet bowl, features auto-generated by GPT-4o would typically include: flush performance, ease of cleaning, comfort, water efficiency, noise level, durability, installation, design/aesthetics, seat quality, and value for money.

Based on the sentiment scoring methodology, flush efficiency and comfort height received high ratings, while durability could not be reliably evaluated due to zero mentions, reflecting broadly positive user sentiment around core functionality. In contrast, features like ease of cleaning and noise level tend to attract more critical commentary, likely scoring lower, indicating a persistent gap between user expectations and the product's current offering.

What this means for design: high-scoring features represent validated design strengths — areas where user expectations are being met or exceeded and should be preserved in future iterations. Low-scoring features represent friction points where user needs are unmet. Ease of cleaning consistently receiving low ratings signals that users find the bowl geometry, rim structure, or surface finish inconvenient to maintain, pointing to a design opportunity around hygiene accessibility and surface material.

6.4 Lessons Learned

Aggregating sentiment scores by averaging smooths out noise from individual outliers — a single highly emotional review does not skew the overall result. However, averaging can also hide extremes: a feature with 60% very positive and 40% very negative mentions might produce a neutral average score of around 3.0★, masking the polarised nature of user opinion. Tracking % Positive and % Negative separately, as the pipeline does, helps surface these cases.

Additionally, the keyword matching approach splits multi-word features into individual words, meaning a sentence only needs to contain one word from the feature name to be counted as a mention. For example, a feature like "ease of cleaning" could match any sentence containing "ease", "of", or

"cleaning" independently. This can introduce false matches and affect the accuracy of feature-level sentiment scores, and is an important limitation to consider when interpreting the results.

7 Requirement Extraction

7.1 How GPT-4o Extracts Requirements

After sentiment analysis is complete, the pipeline passes two inputs to GPT-4o: the feature sentiment summary (including star ratings, mention counts, and example sentences per feature) and a sample of up to 30 raw review texts. GPT-4o is prompted to act as a product designer and generate 10–15 structured, measurable product requirements based on recurring user praise and complaints.

Each requirement follows the format:
[Category] Measurable requirement description

Example extracted requirements for toilet bowls:

1. [Comfort] The toilet bowl should have an elongated shape to enhance user comfort, as it is longer than a round bowl and preferred by most users.
2. [Flush Efficiency] The toilet should clear out solid waste and clean the bowl effectively with one flush, achieving a flush efficiency rating of at least 4.1 stars.
3. [Design Aesthetics] The toilet should feature a sleek, modern design, such as a rimless style, to enhance the overall look of a bathroom.
4. [Price] The retail price of the toilet should be competitive, ideally ranging from $199 to $500, to appeal to budget-conscious consumers.
5. [Ease of Cleaning] The toilet design should facilitate easy cleaning, potentially incorporating self-cleaning features to reduce the need for manual scrubbing.
6. [Bowl Shape] The toilet should offer both elongated and round bowl options to accommodate different bathroom sizes and user preferences.
7. [Water Usage] The toilet should use less water while maintaining effective flushing, aiming for a water usage rating of at least 3.7 stars.
8. [Installation Process] The installation process should be straightforward and not require extensive renovations, with clear instructions provided.
9. [Durability] The toilet should be constructed from high-quality materials to ensure long-lasting durability, even though specific durability ratings are not provided.
10. [Noise Level] The toilet should operate quietly, minimizing noise during flushing to enhance user experience.

7.2 Analysis: Why Structured Requirements Are Useful

Raw sentiment scores tell us *how* users feel about a feature, but not *what to do about it*. A rating of 2.8★ communicates dissatisfaction, but it does not tell a designer whether to increase capacity, improve power efficiency, or introduce fast-charging. Structured requirements bridge this gap by translating diffuse user sentiment into **actionable, testable design targets**.

This step is critical because it shifts the output from analytical (what is wrong) to prescriptive (what must be true). Requirements in the [Category] description format are also directly usable in downstream design processes — they can be fed into a design brief, a user story backlog, or an engineering specification without further interpretation.

The approach also enables prioritisation: requirements derived from features with the most mentions and lowest sentiment scores represent the highest-priority design interventions, since they reflect the most widespread and strongly felt user dissatisfaction.

7.3 Lessons Learned

GPT-4o is effective at converting qualitative language, the imprecise, emotional vocabulary of user reviews — into structured, quasi-quantitative requirements. However, this process introduces two key risks. First, the model may **hallucinate specifics**: if no review explicitly states a number (e.g. "30 hours"), the model may generate a plausible-sounding but unverified figure. All generated requirements should therefore be cross-validated against the source reviews before being treated as authoritative. Second, the quality of requirements is constrained by the quality of the input data — if the review sample is small or skewed toward a particular user type, the requirements will reflect that bias. In future iterations, increasing the review volume and diversifying sources would improve the reliability of extracted requirements.

8 Results & Insights

8.1 Overview of results

A total of 50 reviews, of which 47 contained usable review text and were included and analysed across **10 key product features** using sentiment analysis. The results show variation in user satisfaction across different aspects of toilet bowls, revealing both strong-performing experiential features and weaker performance in maintenance and efficiency-related areas.

8.2 Feature Performance Analysis

8.2.1 Best Performing Features

The highest-rated features include:

- **Comfort height (4.2 ★)**
- **Flush efficiency (4.1 ★)**
- **Design aesthetics (4.1 ★)**

These results indicate that users are generally satisfied with the comfort, overall design, and basic flushing functionality of the toilet bowl.

Insight:
Users place strong value on **ergonomics and visual appeal**, suggesting that toilet bowls are increasingly evaluated not only on functionality but also on comfort and integration into bathroom design.

8.2.2 Moderately Performing Features

Several features fall within a mid-range performance level:

- **Price (3.9 ★)**
- **Ease of cleaning (3.8 ★)**
- **Bowl shape (3.8 ★)**

Insight:
These features indicate **partial satisfaction**, where user expectations are met to some extent but not fully optimised, particularly in areas affecting daily use and maintenance.

8.2.3 Lowest Performing Features

The lowest-rated features include:

- **Water usage (3.7 ★)**
- **Noise level (3.0 ★)**
- **Installation process (3.0 ★, limited mentions)**

Additionally, **durability** recorded no mentions, limiting meaningful evaluation.

Insight:
Lower ratings in these areas suggest that **efficiency, noise, and long-term reliability remain areas of concern**, although limited mentions for some features reduce the reliability of these conclusions.

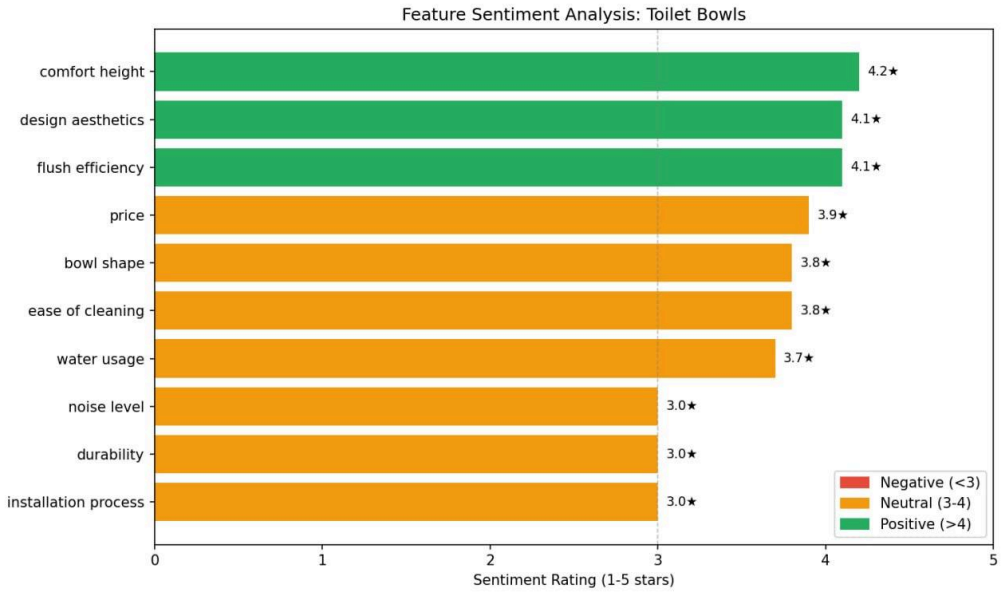

Figure 7: Feature Sentiment Ratings for Toilet Bowls (1–5 scale)

As shown in Figure 7, comfort height, design aesthetics, and flush efficiency achieve the highest ratings, exceeding 4.0 stars. In contrast, features such as noise level, durability, and installation process remain at the neutral threshold of 3.0, indicating weaker performance.

8.3 Patterns Observed

8.3.1 Pattern 1: Functional vs Experiential Gap

Features related to comfort and design perform better than features related to efficiency and maintenance, such as water usage and cleaning. This suggests that current products **prioritise user experience and aesthetics** over functional optimisation and sustainability.

8.3.2 Pattern 2: High Mentions Indicate Unresolved Pain Points

Ease of cleaning (39 mentions) and bowl shape (26 mentions) have the highest discussion frequency but only moderate ratings (~3.8). This indicates that these are critical areas of concern, where users frequently encounter issues that are not fully addressed by existing designs.

8.3.3 Pattern 3: Core Function is Adequate but Not Exceptional

Flush efficiency (4.1) performs relatively well, but its sentiment score (0.563) and mixed feedback suggest that performance may vary across different models or conditions.

8.3.4 Pattern 4: Data Sparsity in Certain Features

Features such as **durability** and **installation process** have very low or zero mentions. This suggests either these aspects are less frequently discussed by users or the dataset is insufficient to capture meaningful insights for these features

8.4 Key Insights

- Users prioritise comfort, ergonomics, and aesthetics, which are currently well-optimised
- Ease of cleaning is a major pain point, with high importance but only moderate satisfaction
- Water efficiency and noise level are underperforming and affect user experience
- Functional performance (flush efficiency) is acceptable but not consistently reliable
- Features with low mention frequency may indicate data limitations rather than strong performance

9 Design Opportunities

9.1 Opportunity 1: Ease of Cleaning

Issue:
Ease of cleaning received only moderate ratings (~3.8) despite having the highest number of mentions (39).

Insight:
Users frequently discuss cleaning, indicating it is a critical aspect of daily use. However, current designs do not sufficiently reduce maintenance effort.

Opportunity:
Enhance cleaning efficiency and reduce user effort through improved surface design and maintenance-friendly features.

9.2 Opportunity 2: Water Efficiency

Issue:
Water usage scored relatively low (3.7), indicating dissatisfaction with efficiency.

Insight:
Users expect toilets to balance strong flushing performance with minimal water consumption.

Opportunity:
Improve water efficiency while maintaining effective flushing performance.

9.3 Opportunity 3: Noise Level

Issue:
Noise level received one of the lowest ratings (3.0).

Insight:
Noise during flushing negatively impacts user experience, especially in shared or residential environments.

Opportunity:
Enhance operational quietness to improve overall user experience.

9.4 Opportunity 4: Installation Process

Issue:
The installation process has low ratings (3.0) and minimal mentions.

Insight:
Although not frequently discussed, installation difficulty can significantly impact initial user experience and product accessibility.

Opportunity:
Improve ease of installation and reduce complexity for users.

9.5 Opportunity 5: Maintenance & Hygiene (Bowl Shape + Cleaning)

Issue:
Bowl shape (3.8) and ease of cleaning (3.8) both show moderate performance.

Insight:
Physical design directly affects hygiene and cleaning effort, making it a key factor in user satisfaction.

Opportunity:
Optimise physical design to improve hygiene and reduce maintenance effort.

9.6 Opportunity Prioritisation

Among the identified opportunities, **ease of cleaning and water efficiency** are prioritised due to their high frequency of mentions and lower satisfaction levels, indicating strong user demand for improvement in these areas.

10 Limitations

10.1 Limited Dataset

The analysis is based on **50 user reviews**, which may not fully represent the diversity of user experiences across different brands and product variations.

10.2 Uneven Feature Representation

Certain features such as **durability and installation process** have very low or zero mentions, limiting the reliability of their sentiment scores.

10.3 Sentiment Interpretation Constraints

The sentiment model may not fully capture nuanced or mixed opinions within reviews, potentially affecting the accuracy of feature-level ratings.

10.4 Aggregation Effects

Averaging sentiment scores across reviews may mask extreme positive or negative experiences, reducing visibility of critical user pain points.

10.5 Language Constraints

The current pipeline is primarily optimised for English-language reviews, which may limit the inclusion and accurate interpretation of reviews written in other languages. As a result, potentially valuable insights from non-English speaking markets may be underrepresented or omitted, reducing the global representativeness of the analysis.

11 Future Improvements

11.1 Expand Dataset Coverage

Future work should include a larger and more diverse dataset from multiple sources to improve representativeness and robustness of findings.

11.2 Improve Feature Coverage

Incorporating additional or dynamically generated features can ensure that all relevant aspects of the product are captured in the analysis.

11.3 Enhance Sentiment Accuracy

Applying more advanced sentiment analysis techniques could improve the detection of nuanced or mixed opinions within user reviews.

11.4 Incorporate Temporal Analysis

Tracking sentiment over time would allow identification of trends and changes in user perception.

12 Lessons Learned

The development of this data-driven analytical pipeline yielded several critical insights regarding the intersection of AI, legal compliance, and design engineering. These lessons serve as a roadmap for future consulting projects utilizing similar text-analytics frameworks.

12.1 API vs. Manual Scraping

A primary technical lesson was the transition from custom scraping scripts to utilizing SerpAPI. While it is possible to build scrapers, doing so often violates the Terms of Service (ToS) of retail platforms. Utilizing SerpAPI acts as a critical compliance buffer, ensuring that our data acquisition respects site-specific crawling policies. We also discovered that major retailers like Amazon and Home Depot employ aggressive anti-bot measures and frequent UI updates that make manual scrapers brittle. Relying on a structured API is a necessity for maintaining a stable, repeatable data pipeline that ensures high uptime and data integrity. This approach also significantly mitigates reputational and legal risks.

12.2 Data Auditability

We learned that exporting data to an excel format (01_raw_reviews.xlsx) immediately after acquisition is vital for Data Auditability. As AI models process and clean text, original nuances can be lost. Having an unedited 'Source of Truth' allows designers to manually verify AI-generated requirements against the unfiltered voice of the customer.

12.3 Preprocessing and Noise

Raw digital text is inherently noisy, containing emojis, HTML tags, and irrelevant shipping complaints. We found that the performance of the RoBERTa sentiment model improved significantly when we focused exclusively on feature-specific sentences, proving that data quality is more impactful than data quantity.

12.4 The Feature Selection Bottleneck

A key analytical takeaway is that the quality of the final report is strictly limited by Stage 3 (Features Identification). If critical product aspects (e.g., Flush Noise or Installation Height) are not identified during this stage, the subsequent sentiment analysis will ignore relevant user complaints. Automation via GPT-4o is efficient, but human-in-the-loop validation is necessary to ensure all technical design parameters are covered.

12.5 Sentence-Level vs. Review-Level Analysis

Standard star ratings often mask specific defects. We learned that sentence-level sentiment mapping is the only way to capture nuanced design opportunities. For example, a user may provide a 4-star rating to compliment the aesthetics while mentioning a frequent clogging issue in the text. Document-level sentiment would label this as positive but our granular approach correctly flags it as a design weakness.

12.6 Efficiency through Automation

By combining specialized NLP models (RoBERTa) with generative AI (GPT-4o), we achieved a level of analysis that would be impossible manually. Processing 50 multi-paragraph reviews across 10 features, a task that would take a human analyst hours, , was completed by the pipeline in minutes.

12.7 Modular and Agnostic Design

One of the most significant project milestones was proving that the pipeline is product-agnostic. By designing the code to initialize with any product query, we created a scalable tool. This taught us that the value of AI in design consulting lies in building flexible frameworks that can be applied to diverse market sectors with minimal reconfiguration.